

Logistic (Function)
(Multiclass Classification)

(x_1) Gpa	(x_2) iq	(y) placement
7.1	71	0
8.5	85	1
9.5	95	2

$\left\{ \begin{array}{ll} \text{yes} & 1 \\ \text{no} & 0 \\ \text{opt-out} & 2 \end{array} \right\}$

Class = 2 $\xrightarrow{\text{softmax}}$ Binary Logistic Regression

Class ≥ 2 $\xrightarrow{\text{softmax}}$ Multiclass Logistic Regression

Function:-

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

K # no. of classes

setup = $\left\{ \begin{array}{l} \text{yes} = 1 \\ \text{no} = 2 \\ \text{opt-out} = 3 \end{array} \right\}$

$\sigma(z)_1 = \text{Probability of } Y_1(1) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$

Softmax

$\sigma(z)_2 =$

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$\sigma(z)_3 = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$

Softmax

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad \text{or} \quad \text{No}(z) = \frac{e^{-z_1}}{e^{-z_1} + e^{-z_2} + e^{-z_3}}$$

$$\sigma(z) = \frac{1}{3} \quad \text{or} \quad \text{optout}(z) = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$\sigma(z_1) + \sigma(z_2) + \sigma(z_3) = 1$$

$$z = \sum_{i=0}^n w_i x_i \quad \text{where } x_i = 1$$

$$z = w_0 + w_1 x_1 + w_2 x_2 = 5 + 6 \cdot (81) + 7 \cdot (10) = 75 + 486 = 561$$

561 $\rightarrow$ Softmax $\rightarrow$ { 0.61, 0.29, 0.10 } $\oplus \rightarrow 1$

Yes No Optout

$\Downarrow$
0.61 (Yes)

iq	cgpa	placement
71	7.1	2
85	8.5	3
110	10	1

Unique class

One Hot Encoding

↓

multiclass
classification
to
binary

classification

$p=1$	$p=2$	$p=3$
0	1	0
0	0	1
1	0	0

↓

m_2

m_3

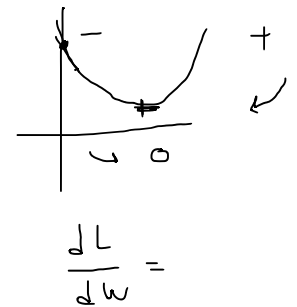
m_1

iq	cgpa	$p=1$
$w_0^{(1)}$	$w_1^{(1)}$	$w_2^{(1)}$

iq	cgpa	$p=2$
$w_0^{(2)}$	$w_1^{(2)}$	$w_2^{(2)}$

iq	cgpa	$p=3$
$w_0^{(3)}$	$w_1^{(3)}$	$w_2^{(3)}$

Weights $\rightarrow$ Random / 1
 $\Downarrow$
 Gradient Descent



$\{7, 70\} \leftarrow$ Prediction

$$z_1 = w_0^{(1)} + 7w_1^{(1)} + 70w_2^{(1)}$$

$$z_2 = w_0^{(2)} + 7w_1^{(2)} + 70w_2^{(2)}$$

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$$z_1 = w_0 + w_1 x_1 + w_2 x_2$$

$$z_3 = w_0^{(3)} + 7w_1^{(3)} + w_2^{(3)}$$

$$\sigma(\text{yes}) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} \quad \bigg| \quad \sigma(\text{No}) = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$\sigma(\text{DD}) = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

Drawback:-

too many Logistic Regression

Cost Function

Normal Logistic Regression:-

✓ binary cross entropy

$$L = -\frac{1}{n} \left[\sum_{i=1}^n y_i \log \hat{y}_i + (1-y_i) \log (1-\hat{y}_i) \right]$$

Softmax Logistic Regression:-

✓ multinomial cross entropy

$$L = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^{\text{class}} \underbrace{y_{ik}^{(i)}}_{\text{circled}} \log (\hat{y}_{ik}^{(i)})$$

x_1	x_2	y	$y_{k=1}$	$y_{k=2}$	$y_{k=3}$
x_{11}	x_{12}	1	1	0	0
x_{21}	x_{22}	2	0	1	0
x_{31}	x_{32}	3	0	0	1

✓

row 1

row 2

$$L = -\frac{1}{n} \left[\underbrace{y_1^1 \log(\hat{y}_1^1)}_{\text{row 1}} + y_2^1 \log \hat{y}_2^1 + y_3^1 \log \hat{y}_3^1 \right. \\ + y_1^2 \log(\hat{y}_1^2) + y_2^2 \log \hat{y}_2^2 + y_3^2 \log \hat{y}_3^2 \quad \text{row 2} \\ + y_1^3 \log \hat{y}_1^3 + y_2^3 \log \hat{y}_2^3 + y_3^3 \log \hat{y}_3^3 + \dots + \\ \left. y_1^m \log \hat{y}_1^m + y_2^m \log \hat{y}_2^m + y_3^m \log \hat{y}_3^m \right]$$

row = m

$$(\quad , \quad , \quad , \quad , \quad , \quad y_3^3 \log \hat{y}_3^3$$

$$L = y_1 \log \hat{y}_1 + y_2 \log \hat{y}_2 + y_3 \log \hat{y}_3$$

$$\hat{y}_1 = \sigma \left(w_1^{(1)} x_{11} + w_2^{(1)} x_{12} + w_0^{(1)} \right)$$

$$\hat{y}_2 = \sigma \left(w_1^{(2)} x_{21} + w_2^{(2)} x_{22} + w_0^{(2)} \right)$$

$$\hat{y}_3 = \sigma \left(w_1^{(3)} x_{31} + w_2^{(3)} x_{32} + w_0^{(3)} \right)$$

$$\begin{bmatrix} w_1^{(1)} & w_2^{(1)} & w_0^{(1)} \\ w_1^{(2)} & w_2^{(2)} & w_0^{(2)} \\ w_1^{(3)} & w_2^{(3)} & w_0^{(3)} \end{bmatrix}$$

$$L = y_1 \log \hat{y}_1 + y_2 \log \hat{y}_2 + y_3 \log \hat{y}_3 + \dots + \underline{y_k \log \hat{y}_k^m}$$

$$= \begin{bmatrix} y_1 & y_2 & y_3 & \dots & y_k^m \end{bmatrix} \log \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_k^m \end{bmatrix}$$

$$\boxed{w_h = w_h - \eta \frac{\partial L}{\partial w}}$$